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LIQUID SUPPLYING APPARATUS

Abstract

There is provided a liquid supplying apparatus including: a liquid container including an internal space, and a tank having a storing chamber. The liquid container has: an upper surface, a nozzle, and a circular wall projecting upward from the upper surface. The tank has: a body; a projecting wall defining a tank recessed part, and a communicating tube positioned in the tank recessed part and having first and second channels which communicate the storing chamber with outside. The circular wall is inserted into the tank recessed part, and the communicating tube is inserted into the opening of the nozzle so as to allow the liquid to flow out from the internal space to the storing chamber. In a case that the liquid container is connected to the tank, the circular wall is positioned inside the projecting wall.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/823,170, filed Aug. 30, 2022, which further claims priority from Japanese Patent Application No. 2021-141929 filed on Aug. 31, 2021. The entire content of both applications is incorporated herein by reference.

BACKGROUND ART

[0002] There is a publicly known ink supplying apparatus wherein an ink is supplied to a tank from a liquid container in a so-called chicken feed system, as a configuration capable of maintaining a liquid surface of the ink stored in the tank to be constant, by supplying the ink successively to the tank from a liquid container connected to the tank every time the ink stored in the tank is consumed.

[0003] In a certain publicly known ink supplying apparatus, the liquid container is connected to the tank from thereabove. The tank is provided with an air introducing part communicating with the atmosphere. The liquid container is provided with an ink outflow pipe and an air inflow pipe. In a state that the liquid container is connected to the tank, the liquid container and the tank are communicated with each other via the ink outflow pipe and the air inflow pipe. In a case that the ink inside the tank is consumed and that the liquid surface of the ink becomes to be lower than a forward end part of the air inflow pipe, the air enters from the air introducing part into the tank, and the air entered into the tank enters into the liquid container via the air inflow pipe. Then, the ink of which amount corresponds to the volume of the air entered into the liquid container is supplied from the liquid container to the tank, via the ink outflow pipe. In a case that the liquid surface of the ink reaches the forward end part of the air inflow pipe, the supply of the ink is stopped. In such a manner, the liquid surface of the ink inside the tank is maintained to be constant.

[0004] In a case that the ink is supplied from the liquid container to the tank, there is such a fear that the ink might adhere to a part, of the tank, at which the liquid container is connected to the tank and that the adhered ink might flow down along the outer circumferential surface of the tank. As a result, there arises such a problem that the outer circumferential surface of the tank and/or the inside of the ink supplying apparatus might be dirtied by the ink.

Description

DESCRIPTION

[0005] The present disclosure has been made in view of the circumstances described above, and an object of the present disclosure is to provide a liquid supplying apparatus wherein a liquid is less likely to flow down along the outer surface of a tank to which a liquid storing liquid container is connected

[0006] According to an aspect of the present disclosure, there is provided a liquid supplying apparatus including: a liquid container in which an internal space configured to store a liquid is

formed; and a tank having a storing chamber configure to store the liquid. The liquid container has: an upper surface, a nozzle projecting upward from the upper surface and having an opening formed in a forward end surface of the nozzle, and a circular wall (an annular wall) which is positioned in a surrounding of the nozzle with a spacing distance from the nozzle, and which projects upward from the upper surface. The tank further has: a body; a projecting wall which projects from an upper surface of the body and which defines a tank recessed part, and a communicating tube which is positioned in the tank recessed part and which has a first channel and a second channel communicating the storing chamber with outside. The circular wall of the liquid container is inserted into the tank recessed part, and the communicating tube of the tank is inserted into the opening of the nozzle of the liquid container to thereby connect the liquid container to the storing chamber of the tank so as to allow the liquid to flow out from the internal space of the liquid container to the storing chamber of the tank. In a connected state in which the liquid container is connected to the tank, the circular wall is positioned at an inner side with respect to the projecting wall.

[0007] In the tank, the liquid accumulated in the tank recessed part does not flow down along the outer circumferential surface of the body of the tank.

[0008] FIG. 1A is a perspective view depicting the outer appearance of a MFP (Multi-Functional Peripheral) 100 in a case that a casing cover 2 is at a shielding position P11, and FIG. 1B is a perspective view depicting the outer appearance of the MFP 100 in a case that the casing cover 2 is at an exposing position P12.

[0009] FIG. 2 is a vertical cross-sectional view schematically depicting the internal structure of a printer part 3.

[0010] FIG. 3A is a perspective view of a tank set 31 in a case that caps 6A to 6D are each at a closing position P21, and FIG. 3B is a plan view schematically depicting the tank set 31, a recording part 32 and the peripheral structure of the recording part 32.

[0011] FIG. 4A is a perspective view of a projecting wall 45, a needle 44 and a receiver key member 46 provided on a body 41, and FIG. 4B is a cross-sectional view of a vertical cross section of the body 41, along a one-dot chain line A-A' and as seen from the orientation of an arrow AR1 of FIG. 4A.

[0012] FIG. 5A is a perspective view of the tank set 31 in a case that the caps 6A to 6D are each at an opening position P22, and FIG. 5B is a perspective view of each of liquid containers 200A to 200D.

[0013] FIG. 6A is a perspective view depicting the outer appearance of a main body 8, and FIG. 6B is an enlarged view of a configuration on a base part 84 of FIG. 6A.

[0014] FIG. 7A is a view of a vertical cross-section of the main body 8, along a one-dot chain line VIIA-VIIA and as seen from the orientation of an arrow AR3 of FIG. 6A, and FIG. 7B is an enlarged view of a configuration on the base part 84 of FIG. 7A.

[0015] FIG. 8A is a perspective view indicating angles D04 to D010 in the receiver key member 46; and FIG. 8B is a view of a vertical cross section of the container cap 9 and a vertical cross section of a part of the main body 8, along a one-dot chain line VIIIB-VIIIB and as seen from the orientation of an arrow AR2 of FIG. 5B.

[0016] FIG. 9A is a perspective view of a tank 4A to which the main body 8 is connected, and FIG. 9B is a vertical cross-sectional view of the main body 8 and the tank 4A, along a one-dot chain line IXB-IXB and as seen from the orientation of an arrow AR4 of FIG. 9A.

[0017] FIG. 10A is a perspective view of a projecting wall 45 and a needle 44 according to a modification, and FIG. 10B is a vertical cross-sectional view of the body 41 as seen from the right side thereof, along a one-dot chain line XB-XB of FIG. 10A.

[0018] FIG. 11A is a perspective view depicting the outer appearance of a main body 8 according to the modification, and FIG. 11B is a plan view of the main body 8 of FIG. 11A in an approaching orientation z22.

[0019] FIG. 12 corresponds to FIG. 8B for depicting a vertical cross-sectional view of the main body **8** and the container cap **9** according to the modification.

[0020] In the following, an embodiment of the present disclosure will be described. Note that the embodiment which is to be explained below is merely an example of the present disclosure; it is needless to say that the embodiment can be appropriately changed without changing the gist of the present disclosure. Further, an up-down direction **z1** is defined, with a posture in which a MFP **100** is installed useably in a horizontal plane (a posture of FIG. 1, also referred to as a “usage posture”), as the reference; a front-rear direction **y1** is defined, with a side on which an opening **1B** of the MFP **100** is provided is defined as a front side (front surface); and a left-right direction **x1** is defined, with the MFP **100** as seen from the front side (front surface). In the present embodiment, in the usable posture, the up-down direction **z1** is the vertical direction, the front-rear direction **y1** and the left-right direction **x1** are parallel to the horizontal plane, and the front-rear direction **y1** and the left-right direction **x1** are orthogonal to each other.

<Configuration of MFP **100**>

[0021] In FIG. 1, the MFP **100** is a multi-functional peripheral, and is provided with a casing **1**, a casing cover **2** and a printer part **3**. The MFP **100** is an example of a “liquid consuming apparatus” and is a part of a liquid supplying apparatus.

[0022] The casing **1** has a substantially rectangular parallelepiped shape and partitions an internal space **1A** (see FIG. 1B) of the MFP **100** from the outside. An upper end of the internal space **1A** is an opening which is oriented upward. An opening **1B** which is oriented frontward is formed in the front surface **11** of the casing **1**, at a location in the vicinity to the center in the left-right direction **x1**. The opening **1B** has a rectangular shape as seen in a plan view from the front, and is communicated with the internal space **1A**.

[0023] The casing cover **2** is coupled or linked to the casing **1**, at a location in the vicinity to an upper rear corner of the casing **1**, with a coupling tool **21** (see FIG. 1B); the casing cover **2** rotates or pivots between a shielding position **P11** (see FIG. 1A) and an exposing position **P12** (see FIG. 1B) about the rotation axis of the coupling tool **21**. At the shielding position **P11**, the casing cover **2** shields constituent components (see FIG. 1B) in the internal space **1A**. The constituent components include a tank set **31**, a recording part **32**, etc. At the exposing position **P12**, the casing cover **2** exposes these constituent components to the outside.

[0024] The casing cover **2** may accommodate, in the inside thereof, a scanner part configured to optically read a manuscript (original). Other than this, the MFP **100** may have a facsimile function, etc.

[0025] In FIG. 2, other than the tank set **31** and the recording part **32** (see FIG. 1B), the printer part **3** is provided with, in the internal space **1A**, a supply tray **33**, a discharge tray **34**, a conveying route **35**, a feeding roller part **36**, a conveying roller part **37**, a discharging roller part **38** and a platen **39**, and records an image on a sheet **S** (paper sheet **S**, paper **S**; see FIG. 2) in the ink-jet recording system.

[0026] The supply tray **33** and the discharge tray **34** are installed in the internal space **1A** via the opening **1B** (see FIGS. 1A and 1i). A plurality of pieces of the sheet **S** are stacked in the supply tray **33**. The discharge tray **34** is positioned above the supply tray **33**, and supports a sheet **S** on which an image is recorded. The conveying route **35** is indicated in FIG. 2 by an arrow of a one-dot chain line, and has a curve part **351** and a straight line (linear) part **352**. The curved part **351** makes a U-turn frontward while extending upward from a rear end of the supply tray **33**. The straight line part **352** extends frontward linearly from a downstream end of the curved part **351** and reaches a rear end of the discharge tray **34**.

[0027] The feeding roller part **36** feeds, one by one, the sheets **S** on the supply tray **33** to an upstream end of the curved part **351**. The conveying roller part **37** is located at a downstream end of the curved part **351**, and feeds the sheet **S** conveyed by the curved part **351** toward the straight line part **352** in a conveying orientation **y2**. The conveying orientation **y2** is frontward in the

straight line part **352**. The discharging roller part **38** is located immediately behind (on the rear side) of the discharge tray **34** in the straight line part **352**, and discharges the sheet S conveyed in the straight line part **352** to the discharge tray **34**.

[0028] The platen **39** is positioned, in the straight line part **352**, between the conveying roller part **37** and the discharging roller part **38**, and supports the sheet S, which is fed out from the conveying roller part **37**, from therebelow. The recording part **32** is positioned above the platen **39**, and is provided with a carriage **321** and a recording head **322**. The carriage **321** moves in a reciprocal manner in a main scanning direction x_2 which is parallel to the left-right direction x_1 . The recording head **322** is mounted on the carriage **321** so that a lower surface of the recording head **322** faces or is opposite to an upper surface of the platen **39**, via the straight line part **352**. A plurality of nozzles **323** are aligned, in the front-rear direction y_1 and the left-right direction x_1 , in the lower surface of the recording head **322**. The recording head **322** discharges or ejects inks of four colors (four color inks) stored inside the recording head **322**, from the plurality of nozzles **323**. The ink(s) is (are) an example of a “liquid”. The four colors are Y color (yellow), M color (magenta), C color (cyan) and K color (black). The recording head **322** discharges the ink(s) from the plurality of nozzles **323** toward the sheet S stopped on the platen **39** while moving together with the carriage **321** in the main scanning direction x_2 at a constant speed. With this, an image corresponding to one pass is recorded on the sheet S. In a case that recording of the image corresponding to one pass is ended, the sheet S is conveyed in the conveying orientation y_2 by a width corresponding to a unit line feed by an intermittent conveyance by the conveying roller part **37**. This image recording and the intermittent conveyance are alternately repeated to thereby record an image on the entirety of the sheet S.

<Tank Set **31**>

[0029] In FIGS. 3A and 3B, the tank set **31** is provided with four tanks **4A** to **4D**, two holding members **51A** and **51B**, four caps **6A** to **6D**, and two tank covers **52A** and **52B**. In FIG. 3B, the holding members **51A** and **51B**, the caps **6A** to **6D** and the tank covers **52A** and **52B** are not illustrated.

[0030] The tanks **4A** to **4D** are installed at a location immediately behind the front surface **11**. The tank **4A** is positioned on the left side with respect to the supply tray **33**. The tanks **4B** to **4D** are positioned on the right side with respect to the supply tray **33**, and are arranged side by side from the left to the right, in an order of the tank **4B**, the tank **4C** and the tank **4D**.

<Tank **4A**>

[0031] The tank **4A** is an example of a “tank”, and is provided with a body **41**. The body **41** has a substantially rectangular parallelepiped shape of which size in the left-right direction x_1 is smaller than a size thereof in the up-down direction z_1 and a size thereof in the front-rear direction y_1 . The body **41** partitions a storing chamber **47** (see FIG. 4B) configured to store the ink of the K color from the outside. The body **41** is formed of an injection molding of a resin material having a light-translucency, etc., except for a side surface, of the body **41**, on one side in the left-right direction x_1 thereof. The side surface on the one side in the left-right direction x_1 is closed or sealed by a resin film which is thinner than other parts different from the side surface on the one side in the left-right direction x_1 .

[0032] As depicted in FIG. 3B, one end of one of tubes **42** made of a flexible resin is connected to a location, of the body **41**, in the vicinity of the rear end of the body **41**. The other end of one of the respective tubes **42** is connected to the recording head **322**. In accordance with consumption of the ink inside the recording head **322**, the ink inside the body **41** is supplied to the recording head **322** via one of the tubes **42**. An atmosphere communicating hole is also formed at a location, of the body **41**, in the vicinity of the rear end of the body **41**.

[0033] In FIG. 4A, the body **41** has an upper surface **43** which is parallel to a horizontal plane. A needle **44** (an example of a “communicating tube”), a projecting wall **45** and a receiver key member **46** are integrally provided on the upper surface **43**.

[0034] The needle **44** is a member which has a shape of a circular pipe and which is elongated in the up-down direction **z1**. An outer circumferential surface and an inner circumferential surface of the needle **44** share an axis **Ax1** which is parallel to the up-down direction **z1**. The needle **44** extends upward perpendicularly from the upper surface **43**, and extends also downward perpendicularly with respect to the upper surface **43** and toward the inside of the storing chamber **47**. An upper end of the needle **44** is located above the receiver key member **46**. A lower end of the needle **44** is located above a bottom surface **47A** of the storing chamber **47**, as depicted in FIG. **9B**. In FIGS. **4A** and **4B**, the needle **44** has two channels **441** and **442**, and a partition wall **443**. The two channels **441** and **442** extend downward linearly from a location, in the needle **44**, which is lower to some extent from the upper end of the needle **44**, toward the storing chamber **47** of the body **41**. An upper end of each of the channels **441** and **442** is released upward, and a lower end of each of the channels **441** and **442** is released downward. The channels **441** and **442** are partitioned with respect to each other by the partition wall **443** which expands in the up-down direction **z1** and the left-right direction **x1** in an entire area between the upper end and the lower end of the needle **44**. The partition wall **443** extends up to a location above the channels **441** and **442**. The channel **441** extends up to a location below the channel **442**. An opening in the upper end of the channel **441** is an example of an “inlet port”. The channels **441** and **442** are examples of a “first channel” and a “second channel”, respectively.

[0035] In FIG. **4A**, the projecting wall **45** has curved plates **71** and **72** and connecting plates **73** and **74** so as to define an area which is elliptic in a plan view thereabove (hereinafter referred also to as a “first plan view”) in the upper surface **43**. The projecting wall **45** projects from the upper surface **43** upward only by a distance **Dz1** (see FIG. **4B**). The distance **Dz1** is an example of a “second distance”. A projecting end surface (namely, an upper end surface) of the projecting wall **45** is parallel to the upper surface **43**.

[0036] The curved plates **71** and **72** face each other and are apart from each other in the front-rear direction **y1** with the needle **44** being interposed therebetween. Each of the curved plates **71** and **72** is substantially semi-cylindrically shaped in the first plan view. Inner circumferential surfaces of the curved plates **71** and **72** are each circular-arc shaped, while being rotationally moved approximately 180 degrees from each other about the axis **Ax1**. The curved plate **71** is positioned in front of the curved plate **72** and swells or protrudes frontward.

[0037] The connecting plates **73** and **74** face each other and are apart from each other in the left-right direction **x1** with the needle **44** being interposed therebetween. A right side surface of the connecting plate **73** and a left side surface of the connecting plate **74** face each other and are apart from each other in the left-right direction **x1** with the needle **44** being interposed therebetween, and are symmetrically shaped to each other, with a virtual plane including the axis **Ax1** and parallel to the up-down direction **z1** and the front-rear direction **y1** as the reference. Each of the right side surface of the connecting plate **73** and the left side surface of the connecting plate **74** is substantially rectangular-shaped which is elongated in the front-rear direction **y1**. A distance in the left-right direction **x1** between the connecting plates **73** and **74** is not less than the diameter of an outer circumferential surface **853** (see FIG. **6B**) of a circular wall **85** (which will be described later on), as appreciated also from FIG. **9A**. The connecting plate **73** connects or links left ends, respectively, of the curved plates **71** and **72** to each other, and the connecting plate **74** connects or links right ends, respectively, of the curved plates **71** and **72** to each other.

[0038] The receiver key member **46** projects upward in the upper surface **43**, perpendicularly between the needle **44** and the projecting wall **45**. The receiver key member **46** partitions or defines a key hole **48** (an example of a “tank recessed part”) which is released upward, together with the upper surface **43** and the projecting wall **45**. Although the key hole **48** is in conformity to a key member **88** (see FIG. **6B**) formed in the liquid container **200A**, the key hole **48** is not in conformity to a key member of each of the liquid containers **200B** to **200D** different from the liquid container **200A**. The receiver key member **46** includes a base part **461** and ribs **462** to **4610**.

[0039] The base part **461** is substantially cylindrical shape or substantially ringed shape in the first plan view (see FIG. 4A), and projects from the upper surface **43** perpendicularly up to a location below the distance $Dz1$ (see FIG. 4B). The diameter of the inner circumferential surface of the base part **461** is not less than the inner diameter of a circular flat surface **862B** (see FIG. 6B) and less than the outer diameter of the circular flat surface **862B**. The diameter of the outer circumferential surface of the base part **461** is substantially same as the outer diameter of the circular flat surface **862B**. The upper surface of the base part **461** is parallel to the upper surface **43** and is oriented upward, and serves as a contact surface **461A** with which the circular flat surface **862B** is brought into contact at the time of ink replenishment.

[0040] Each of the ribs **462** and **463** extends linearly in the left-right direction $x2$ between the needle **44** and the connecting plate **73**. In each of the ribs **462** and **463**, a part near to a right end thereof projects upward perpendicularly from the contact surface **461A**, and another part different from the part near to right end projects upward perpendicularly from the upper surface **43**. Each of the ribs **462** and **463** are apart from the needle **44** leftward, and are apart from the connecting plate **73** rightward. The ribs **462** and **463** are symmetrically shaped to each other, with a virtual plane including the axis $Ax1$ and parallel to the up-down direction $z1$ and the left-right direction $x1$ as the reference; and the ribs **462** and **463** are positioned with a gap which is substantially constant in the front-rear direction $y1$ therebetween.

[0041] The ribs **464** and **465** have shapes, respectively, which are obtained by rotationally moving the ribs **462** and **463**, respectively, about the axis $Ax1$ by approximately 180 degrees only.

[0042] A rib **466** extends linearly in the front-rear direction $y1$ between the needle **44** and the curved plate **72**. In the rib **466**, a part near to a front end thereof projects upward perpendicularly from the contact surface **461A**, and another part different from the part near to the front end projects upward perpendicularly from the upper surface **43**. The rib **466** is apart from the needle **44** rearward, and is apart from the curved plate **72** frontward. The rib **466** is positioned at a location which is intermediate between the ribs **463** and **464** in a circumferential direction $\theta1$ of the axis $Ax1$. The width in the left-right direction $x1$ of the rib **466** is substantially constant over an entire area thereof in the front-rear direction $y1$.

[0043] A rib **467** is substantially semi-cylindrical shaped in the first plan view, and projects upward perpendicularly from the upper surface **43**. The rib **467** is positioned at a substantially intermediate location between the needle **44** and the curved surface **71** in a radial direction $r1$ of the axis $Ax1$. The rib **467** extends substantially parallel to the curved plate **71** along the outer circumferential surface of the base part **461**, within a range between both inner ends, respectively, of the ribs **462** and **465**, in the circumferential direction $\theta1$. The term “inner end” is an end in an orientation $r12$ approaching toward the axis $Ax1$ in the radial direction $r1$ (hereinafter referred also to as an “inward orientation” $r12$).

[0044] Upper end surfaces, respectively, of the ribs **462** and **467** are flush with one another, and are located above the projecting wall **45**. The term “flush” means that there is no difference in height, and is parallel.

[0045] Each of ribs **468** and **469** has a substantially quarter-cylindrical shape which is coaxial with the axis $Ax1$ in the first plan view, and projects upward perpendicularly from the upper surface **43**. An upper end surface of each of the ribs **468** and **469** is parallel with respect to the horizontal plane over an entire area thereof in the circumferential direction $\theta1$. The ribs **468** and **469** are positioned, in the radial direction $r1$, at a substantially intermediate location between the curved plate **72** and the base part **461**. The rib **468** extends substantially parallel to the base part **461** and the curved plate **72** in the circumferential direction $\theta1$ at a location between both outer ends, respectively, of the ribs **463** and **466**. The rib **469** extends substantially parallel to the base part **461** and the curved plate **72** in the circumferential direction $\theta1$ at a location between both outer ends, respectively, of the ribs **466** and **464**. The term “outer end” is an end in an orientation $r11$ which is opposite to the inward orientation $r12$ (hereinafter referred also to as an “outward orientation” $r11$). In FIG. 4A, an

example of each of the radial direction **r1**, the outward orientation **r11** and the inward orientation **r12**.

[0046] The upper end surface of the rib **468** is flush with the upper end surface of the rib **463** within a range of an angle **D01** from the rib **463** in the circumferential direction **01**. The upper end of the rib **468** is positioned at a location which is lower than the upper end surface of the rib **463** and is positioned above the contact surface **461A** in the up-down direction **z1** outside the range of the angle **D01**. Namely, a cutout **468A** is formed in the rib **468** outside the range of the angle **D01**.

[0047] The upper end surface of the rib **469** is flush with the upper end surface of the rib **466** within a range of an angle **D02** from the rib **466** in the circumferential direction **01**. The upper end surface of the rib **469** is positioned at a location which is lower than the upper end surface of the rib **466** and is positioned above the contact surface **461A** in the up-down direction **z1** outside the range of the angle **D02**. Namely, a cutout **469A** is formed in the rib **469** outside the range of the angle **D02**.

[0048] The rib **4610** projects outward in the outward orientation **r11** on the outer circumferential surface of the rib **467**, from a position which is apart from the rib **465** only by an angle **D03** in the circumferential direction **01**. A projecting end of the rib **4610** is apart from the curved plate **71** in the inward orientation **r12**. The rib **4610** extends perpendicularly from the upper surface **43** up to a position below the upper end surface of the rib **467**.

[0049] Although the rib **4610** projects outward in the outward orientation **r11** on the outer circumferential surface of the rib **467**, from the position which is apart from the rib **465** only by the angle **D03** in the circumferential direction **01**, the rib **4610** is inclined with respect to the front-rear direction **y1**. An outer end of the rib **4610** is apart from the curved plate **71** in the inward orientation **r12**. An upper end surface of the rib **4610** is parallel to the upper end surface of the rib **467**, at a location below the upper end surface of the rib **467**. The width in the circumferential direction **01** of the rib **4610** is substantially constant over the entire area thereof in the radial direction **r1**.

[0050] Inner ends, respectively, of the ribs **462** to **466** and an inner circumferential surface of the rib **467** define, together with the outer circumferential surface of the needle **44** and the contact surface **461A**, the outer shape of a cylindrical space **46A** which is opened upward. At a time of ink replenishment, a small diameter part **862** having a cylindrical shape (see FIG. **6B**) is inserted to the cylindrical space **46A**.

[0051] The upper surface **43**, the outer circumferential surface of the needle **44**, and the inner circumferential surface of the base part **461** define a circular space **46B**. The circular space **46B** is circular in the first plan view, and is recessed downward with respect to the contact surface **461A** in the surrounding of the needle **44**. A circular flat surface **862C** and a circular inclined surface **862D** in the small diameter part **862** (see FIG. **6B**) enter into the circular space **46B**, at a time of ink replenishment.

[0052] Facing surfaces, respectively, of the ribs **463** and **466** in the circumferential direction **01**, the inner circumferential surface of the rib **468**, the outer circumferential surface of the base part **461** and the upper surface **43** define a partially circular space **46C**. Facing surfaces, respectively, of the ribs **466** and **464** in the circumferential direction **01**, the inner circumferential surface of the rib **469**, the outer circumferential surface of the base part **461** and the upper surface **43** define a partially circular space **46D**. Each of the partially circular spaces **46C** and **46D** has a substantially quarter-circular shape in the first plan view, and is recessed downward from the contact surface **461A**.

[0053] The upper surface **43**, the inner circumferential surface of the projecting wall **45** and the ribs **462** to **4610** define an external space **46E**. Apart in the vicinity of the forward end part of the circular wall **85** (see FIGS. **6A** and **6B**) is positioned in the external space **46E** at the time of ink replenishment. The external space **46E** communicates with the circular space **46A** via a gap between the ribs **462** and **463** and a gap between the ribs **464** and **465**.

<Tanks 4B to 4D>

[0054] In FIGS. 3A and 3B, each of the tanks 4B to 4D is another example of the “tank”, and is provided with a body which is configured similarly to the body 41, except for the following points. In the body of each of the tanks 4B to 4D, a receiver key member having a function similar to that of the receiver key member 46 is formed. The receiver key member of each of the tanks 4B to 4D defines a key hole opened upward by a combination of a plurality of ribs, together with an upper surface and a projecting wall. The respective receiver key members of the tanks 4B to 4D have three-dimensional shapes which are mutually different among the tanks 4B to 4D, and which are also different from that of the receiver key member 46 of the tank 4A. The three-dimensional shape of each of the receiver key members is determined by the sizes and/or positions in the left-right direction x1, the front-rear direction y1 and the up-down direction z1 of the respective ribs, or by the number (quantity) of the ribs. The bodies of the respective tanks 4B, 4C and 4D are different from the body 41 in view of a point of storing the inks of C color, M color and Y color, respectively. The bodies of the respective tanks 4B to 4D may also be different from the body 41 in view of the volume of the ink.

<Holding Members 51A, 51B>

[0055] In FIG. 3A, the holding member 51A covers the upper surface 43 (see FIG. 4A) of the body 41. The holding member 51A is formed with a through hole 511 (see FIG. 5A) through which the projecting wall 45 and the needle 44 (see FIG. 4A) are inserted. The holding member 51B collectively covers the upper surfaces of the tanks 4B to 4D (see FIG. 3B). The holding member 51B is formed with through holes 511B to 511D (see FIG. 5A).

[0056] Cylindrical walls and needles possessed by the tanks 4B to 4D are inserted through the through holes 511B and 511D, respectively.

[0057] A bearing 53A is provided on the holder member 51A, at a location behind the through hole 511A. Bearings 53B to 53D are provided on the holder member 51B, each at a location behind one of the through holes 511B to 511D. Each of the bearings 53A to 53D has the rotation axis which is parallel to the left-right direction x1, and supports one of the caps 6A to 6D between a closing position P21 (see FIG. 3A) and an opening position P22 (see FIG. 5A), rotatably about the rotation axis of one of the bearings 53A to 53D.

<Caps 6A to 6D>

[0058] In FIGS. 3A and 3B and FIGS. 5A and 5B, the cap 6A has a rubber part 61A and an arm part 62A. The rubber part 61A has a cylindrical shape of which diameter is greater than that of the needle 44 (see FIGS. 4A and 4B), and has a hole through which the needle 44 is inserted. Note that FIG. 5A does not illustrate the needle 44, for the sake of convenience. The arm part 62A is formed of a resin material which is harder than that forming the rubber part 61A, and has an elongated stick or bar-like shape. The rubber part 61A is attached to one end of the arm part 62A. The other end of the arm part 62A is provided with a rotation shaft through which the bearing 53A is inserted.

[0059] As depicted in FIG. 3A, in a case that the cap 6A is at the closing position P21, the arm part 62A extends frontward from the bearing 53A, and the rubber part 61A fits (is inserted) into the key hole 48 via the through hole 511A of the holding member 51A. In this situation, the needle 44 is inserted into the hole of the rubber part 61A. Note that for the sake of convenience, FIG. 5A does not illustrate the needle 44 and the key hole 48. With this, any leaking and/or drying of the ink inside the body 41 is/are prevented. The opening position P22 is a position which is rotated, from the closing position P21, at an angle approximately in a range of 90 degrees to 100 degrees about the rotation axis of the bearing 53A.

[0060] Although the caps 6B to 6D have a similar configuration to that of the cap 6A, the caps 6B to 6D are different from the cap 6A in view of that each of the caps 6B to 6D loosely fits to the key hole provided on one of the liquid containers 200B to 200D (see FIG. 3B, etc.) via one of through holes 511B to 511D of the holding member 51B.

<Tank Covers 52A, 52B>

[0061] In a case that the casing cover 2 is at the exposing position P12 (see FIG. 1B), each of the tank covers 52A and 52B is rotatable (pivotable), about the rotation axis located on the rear side of one of the bearings 53A to 53D, between a covering position P31 and an exposing position P32 (see FIG. 3A). In a case that the tank cover 52A is at the covering position P31, the tank cover 52A covers the holding member 51A, the cap 6A and the bearing 53A from thereabove. In a case that the tank cover 52B is at the covering position P31, the tank cover 52B covers the holding member 51B, the caps 6B to 6D and the bearings 53B to 53D from thereabove. The exposing position P32 is a position which is rotated, from the covering position P31, at an angle approximately in a range of 90 degrees to 100 degrees about the rotation axis of each of the tank cover 52A and the tank cover 52B.

<Liquid Containers 200A to 200D>

[0062] As depicted in FIG. 5B, in the MFP 100 (see FIGS. 1A and 1), for example, four liquid containers 200A to 200D are used for replenishing the tanks 4A to 4D with the inks, respectively. The liquid containers 200A to 200D are a remaining part of the liquid supplying apparatus. Note that in FIG. 5B, the liquid container 200A is illustrated to be greater than the liquid containers 200B to 200D, for the sake of convenience. The liquid container 200A stores a replenishing ink for the tank 4A (the ink of K color). The liquid container 200A is provided with a main body 8 and a container cap 9. The liquid container 200A is an example of a “liquid storing bottle”, and the container cap 9 is an example of a “cap”.

<Main Body 8>

[0063] In FIG. 6A, the main body 8 has a bottom part 81, a trunk part 82, a shoulder part 83, a base part 84, a circular wall 85 and a neck part 86.

<Bottom Part 81>

[0064] The bottom part 81 is a flat part of a bottom wall which has a substantially disc-shape. A posture assumed by the main body 8 in a case that the bottom part 81 is brought into contact with a horizontal plane 300 (see FIG. 5B) to thereby place the main body 8 on the horizontal plane 300 is referred to as a “placement posture”. In the following explanation, unless otherwise noted, the posture of the main body 8 is the placement posture. A virtual line passing through the center of the bottom part 81 and orthogonal to the bottom part 81 is defined as an “axis Ax2”. In an axial direction z2 in which the axis Ax2 extends, an orientation from the bottom part 81 toward the neck part 86 is also referred to as a separating orientation z21, and an orientation reverses to the separating orientation is also referred to as an approaching orientation z22. In a radial direction r1 of the axis Ax2, an orientation approaching toward the axis Ax2 is also referred to as an inward orientation r21, and an orientation reverse to the inward orientation r21 is also referred to as an outward orientation r22. FIGS. 5A, 5B, 6A and 6B depict only an example of each of the radial direction r2, the inward orientation r21 and the outward orientation r22.

<Trunk Part 82, Shoulder Part 83>

[0065] The trunk part 82 is a wall which has a substantially cylindrical shape extending from an outer edge of the bottom part 81 in the separating orientation z21. The shoulder part 83 is a wall which extends from an extending end of the trunk part 82 in the inward orientation r21. The shoulder part 83 is inclined with respect to the radial direction r2 of the axis Ax2 so as to further apart from the bottom part 81 as approaching closer to the axis Ax2. An extending end of the shoulder part 83 is away (apart) from the axis Ax2 in the outward orientation r22, and has a circular shape in a plan view from the approaching orientation z22 (hereinafter also referred to as a “second plan view”).

<Base Part 84>

[0066] The base part 84 has a side wall and an upper wall. The side wall projects from the extending end of the shoulder part 83 in the separating orientation z21 (namely, upward) and has a substantially cylindrical shape which is substantially coaxial with the axis Ax2. The upper wall projects from the projecting end (namely, the upper end) of the side wall of the base part 84 in the

inner orientation **r21** and has a substantially ringed shape in the second plan view. In an upper wall of the base part **84**, an upper surface **841** (an example of an “upper surface”) defines an upper end of the base part **84**, and is a surface parallel to the bottom part **81**.

<Storing Chamber **87**>

[0067] As depicted in FIGS. 7A and 7B, the main body **8** has a space which is defined by the bottom part **81**, the trunk part **82**, the shoulder part **83** and the base part **84**, as a storing chamber **87**. The ink of the K color which is to be supplied to the tank **4A** and with which the tank **4A** is to be replenished is stored in the storing chamber **87**. The storing chamber **87** is an example of an “internal space” of the bottle body.

<Circular Wall **85**>

[0068] In FIGS. 6A, 6B, 7A and 7B, the circular wall **85** has a bottomed cylindrical shape which is coaxial with the axis **Ax2**. In the circular wall **85**, an end in the approaching orientation **z22** is sealed (see FIG. 7A) so as to form a bottom surface **851** which is ring-shaped (which has an annular shape) in the second plan view. The bottom surface **851** is a surface which is positioned in the separating orientation **z21** with respect to the upper surface **841**, which is parallel to the upper surface **841** and which is oriented (faces) upward. The circular wall **85** projects in the separating orientation **z21** from an outer edge of the bottom surface **851**, and extends up to a position which is apart, with the upper surface **841** as the reference, only by a distance **Dz2** (an example of a “first size”) in the separating orientation **r21** (see FIG. 4B). The distance **Dz2** is longer than the distance **Dz1**. The separating orientation **r21** is upward in the placement posture. The circular wall **85** has an end surface **852** at an end thereof in the separating orientation **z21**. The end surface **852** is ring-shaped in the second plan view, and expands parallel to the upper surface **841**. The end surface **852** surrounds an opening in the separating orientation **z21** of the circular wall **85**. A width in the radial direction **r2** of the circular wall **85** is substantially constant over the entire area thereof in a circumferential direction **62**, and is not more than a distance in the left-right direction **x1** between the rib **462** and the connecting plate **73** (see FIG. 4A). With the above-described configuration, the circular wall **85** is insertable to a space between the receiver key member **46** and the projecting wall **45** in the external space **46E** (see FIG. 4A), and the end surface **852** is capable of making contact with the upper surface **43** in the external space **46E** (see FIG. 4A).

<Neck Part **86**>

[0069] In FIGS. 6A, 6B, 7A and 7B, the neck part **86** is an example of a “nozzle”, and has a large diameter part **861** and a small diameter part **862**.

[0070] The large diameter part **861** is a substantially cylindrical body having an outer circumference surface **861A** and a circular flat surface **861B**. The outer circumferential surface **861A** extends from the bottom surface **851** in the separating orientation **z21**, and projects up to a location above the upper surface **841**. The outer circumferential surface **861A** is apart from the circular wall **85** in the inward orientation **r21** over the entire area thereof in the circumferential direction **62**. The circular flat surface **861B** extends from the projecting end of the outer circumferential surface **861A** in the inward orientation **r21** by a substantially constant distance. The circular flat surface **861B** is ring shaped in the second flat view, and is substantially parallel to the upper surface **841**.

[0071] The small diameter part **862** is a substantially cylindrical body having an outer circumference surface **862A**, circular flat surfaces **862B**, **862C** and a circular inclined surface **862D**. The outer circumferential surface **862A** extends from an inner edge of the circular flat surface **861B** in the separating orientation **z21**, and has a diameter smaller than that of the outer circumferential surface **861A**. The circular flat surface **862B** is ring-shaped in the second plan view, and extends from a projecting end of the outer circumferential surface **862A** in the inward orientation **r21** by a substantially constant distance and substantially in parallel to the upper surface **841**. The circular flat surface **862C** is a forward end surface in the separating orientation **z21** of the neck part **86**, and is ring-shaped in the second plan view. The circular flat surface **862C** is an

example of a “forward end surface”, and is connected to the circular flat surface **862B** via the circular inclined surface **862D** at a position which is apart from the circular flat surface **862B** in the inward orientation **r21** and in the separating orientation **z21**.

[0072] The neck part **86** defines a channel **862F** via which the ink stored in the storing chamber **87** passes. As depicted in FIGS. 7A and 7B, the channel **862F** is continuous to the storing chamber **87** at an end in the approaching orientation **z22** of the channel **862F**, and is continuous to an outflow port **862E** (an example of an “opening”) which is formed in the circular flat surface **862C**, at an end in the separating orientation **z21** of the channel **862F**. In the channel **862F**, a part surrounded by the large diameter part **861** has a diameter larger than a diameter of a part, of the channel **862F**, which is surrounded by the small diameter part **862** (see FIG. 7B). The outflow port **862E** is circular shaped (round shaped) in the second plan view. The outflow port **862E** has a diameter which is slightly larger than the needle **44** (see FIGS. 4A and 4B), and allows the ink flowing through the channel **862F** to flow to the outside of the liquid container **200A**.

[0073] In the present embodiment, the outer circumferential surfaces **861A** and **862A**, the circular flat surfaces **861B**, **862B** and **862C**, the circular inclined surface **862D** and the outflow port **862E** are coaxial with the axis **Ax2**. However, the present embodiment is not limited to or restricted by this; it is allowable that the axis of at least one of the outer circumferential surfaces **861A** and **862A**, the circular flat surfaces **861B**, **862B** and **862C**, the circular inclined surface **862D** and the outflow port **862E** is not coaxial with the axis **Ax2**. In the present embodiment, the outflow port **862E** is round shaped in the second plan view. However, the present disclosure is not limited to this; the outflow port **862E** may have a shape which is different from the round shape in the second plan view.

<Relationship Between Circular Wall **85** and Neck Part **86**>

[0074] In FIGS. 7A and 7B, the circular wall **85** is positioned in the surrounding of the neck part **86**, with a spacing distance from the neck part **86** in the outward orientation **r22**, and defines a cylindrical space **86A** between the circular wall **85** and the neck part **86**. In the separating orientation **z21**, a forward end of the circular wall **85** is apart from the bottom part **81** farther than the forward end of the neck part **86**. In the placement posture, the forward end (namely, the upper end) of the circular wall **85** is positioned above the circular flat surface **862B** (namely, the forward end surface) of the neck part **86**.

<Key Member **88**, First Rib **881**, Second Rib **882** and Third Rib **883**>

[0075] In FIG. 6B, the main body **8** is further provided with a key member **88**. The key member **88** projects perpendicularly from the upper surface **841**, the bottom surface **851** and the circular flat surface **861B** in the separating orientation **z21**, at a position between the circular wall **85** and the small diameter part **862**. In the placement posture, an upper end of the key member **88** is closer to the upper surface **841** than the upper end of the circular wall **85**. The key member **88** includes a first rib **881**, a second rib **882** and a third rib **883** which are in conformity to or are loosely in conformity to the receiver key member **46**.

[0076] The first rib **881** connects to each of the small diameter part **862** and the circular wall **85**. As the first rib **881**, three first ribs **881A** to **881C** are exemplified in FIG. 6B. Each of the first ribs **881A** to **881C** projects perpendicularly from the bottom surface **851** and the circular flat surface **861B** in the separating orientation **z21**. An inner end and an outer end of each of the first ribs **881A** to **881C** are integral, respectively, to the small diameter part **862** and the circular wall **85**. A projecting end of each of the first ribs **881A** to **881C** expands, in the axial direction **z2**, substantially parallel to the upper surface **841** at a position closer to the end surface **852** of the circular wall **85** than the circular flat surface **861B**.

[0077] The first ribs **881A** and **881B** have shapes, respectively, which are rotationally moved from each other by approximately 180 degrees in the circumferential direction $\theta 2$ of the axis **Ax2**. The projecting end surface of each of the first ribs **881A** and **881B** is flush with the circular flat surface **862B**. At the time of ink replenishment, the first rib **881A** has a size which is insertable to a gap

between the ribs **464** and **465** (see FIG. 4A) from thereabove; and the first rib **881B** has a size which is insertable to a gap between the ribs **462** and **463** (see FIG. 4A) from thereabove.

[0078] The first rib **881C** is apart only by an angle $D04$ in an orientation $\theta21$ in the clockwise orientation in the circumferential direction $\theta2$, with the first rib **881A** as the reference, in the second plan view. The angle $D04$ is greater than angles $D05$ angle $D06$ which will be described later on. A projecting end surface of the first rib **881C** is positioned, in the axial direction $z2$, to a slight extent in the approaching orientation $z22$ with respect to the circular flat surface **862B**, and defines a stepped part (a difference in height) with respect to the circular flat surface **862B**. The first rib **881C** has a size fittable to the cutout **468A** (see FIG. 4A) from thereabove at the time of ink replenishment.

[0079] Although the second rib **882** connects to the small diameter part **862**, the second rib **882** does not connect to the circular wall **85**. In FIG. 6B, one piece of a second rib **882A** is exemplified as the second rib **882**. The second rib **882A** extends linearly, in the second plan view, in the outward orientation $r22$ along the circular flat surface **861B** from a position, in the outer circumference surface **862A**, which is apart only by an angle $D05$ (see The. 8A) in the orientation $\theta21$, with the first rib **881A** as the reference. The second rib **882A** extends linearly, in the second plan view, in the outward orientation $r22$ along the circular flat surface **861B** also from a position, in the outer circumference surface **862A**, which is apart only by an angle $D06$ (see FIG. 8A) in the orientation $\theta21$ from the first rib **881A**. The angle $D05$ is greater than 0 (zero), and the angle $D06$ is greater than the angle $D05$. The second rib **882A** extends in a circular-arc shape along an outer edge of the circular flat surface **861B** between the angles $D05$ and $D06$. Although the respective projecting end surfaces in the separating orientation $z21$ of the second rib **882A** expand substantially parallel to the upper surface **841**, the respective projecting end surfaces are positioned in the approaching orientation $z22$ to some extent with respect to the circular flat surface **862B**, and define a stepped part (difference in height) with respect to the circular flat surface **862B**. The width of the second rib **882A** is substantially constant over an entire area in an extending direction in which the second rib **882A** extends. The second rib **882A** has a size loosely fittable to the partially circular space **46D** (see FIG. 4A) at the time of ink replenishment.

[0080] Although the third rib **883** does not connect to the small diameter part **862**, the third rib **882** connects to the circular wall **85**. In FIG. 6B, one piece of a third rib **883A** is exemplified as the third rib **883**. The third rib **883A** is positioned in the outward orientation $r22$ to be apart with respect to the small diameter part **862**. The third rib **883A** extends in a circular arc shape, in the second plan view, along the outer edge of the circular flat surface **861B** within a range from an angle $D07$ (see FIG. 8A) up to an angle $D08$ (see FIG. 8A) in a counterclockwise orientation in the circumferential direction $\theta2$, with the first rib **881A** as the reference. The angle $D07$ is greater than 0 (zero), and the angle $D08$ is greater than the angle $D07$. The third rib **883A** extends in the outward orientation $r22$ from the both ends in the circumferential direction $\theta2$ toward the circular wall **85**, and connects to the circular wall **85**. A projecting end surface in the separating orientation $z21$ of the third rib **883A** expands parallel to the upper surface **841**, and is positioned in the approaching orientation $z22$ with respect to the end surface **852**. The projecting end surface has a stepped part (difference in height) within a range from an angle $D09$ (see FIG. 8A) up to an angle $D010$ (see FIG. 8A) in the orientation $\theta22$, with the first rib **881** as the reference. The angle $D09$ is greater than the angle $D07$. The angle $D010$ is greater than the angle $D09$ and is smaller than the angle $D08$. The width of the third rib **883A** is substantially constant over an entire area in an extending direction in which the third rib **883A** extends. The third rib **883A** has a size loosely fittable to a space between the ribs **462** and **4610** in the external space **46E**, at the time of ink replenishment.

<Male Screw **854**, Recessed Parts **855A**, **855B**>

[0081] In FIGS. 6B, 7A and 7B, the main body **8** has a male screw **854** in the outer circumferential wall **853** of the circular wall **85**. The male screw **854** projects in the outward orientation $r22$ from

the outer circumferential surface **853** of the circular wall **85**. The male screw **854** has a helical shape which moves in the outer circumferential surface of the circular wall **85**, in the approaching orientation **z22** at a position which is apart from the end surface **852** in the approaching orientation **z22** only by a distance **Dz3** (see FIG. **4B**), while rotating about the axis **Ax2**. The distance **Dz3** is an example of a “third size”, and is longer than the distance **Dz1**. Recessed part **855A** and **855B** (each an example of a “recessed part”) are formed in the male screw **854**. The recessed part **855A** is formed by cutting out, in the male screw **854**, at least a part thereof in the outward orientation **r22** with respect to the first rib **881A**. The recessed part **855A** is recessed toward the first rib **881A** with respect to a virtual line **c1** (see a broken line in FIG. **6B**) which is obtained by virtually extending an apex of the male screw **854**.

[0082] As depicted in FIG. **8A**, positions which are apart, to some extent, from the first rib **881A** respectively in the orientation $\theta 21$ and $\theta 22$ are defined as positions **P41** and **P42**, respectively. Positions which are apart from the positions **P41** and **P41** respectively in the orientation $\theta 21$ and $\theta 22$ are defined as positions **P51** and **P52**, respectively. A line linking the position **P41** at the foot of the screw thread of the male screw **854** and the position **P51** at the apex of the screw thread is defined as a line **L11**. A line linking the position **P42** at the foot of the screw thread of the male screw **854** and the position **P52** at the apex of the screw thread is defined as a line **L12**. In the second plan view, by cutting out a part, in the male screw **854**, which is surrounded by the outer circumferential surface **853**, the virtual line **c1** and the lines **L11** and **L12**, the recessed part **855A** is formed.

[0083] The recessed part **855B** has a shape which is obtained by rotating the recessed part **855A** by 180 degrees in the circumferential direction $\theta 2$.

<Valve Mechanism **88**>

[0084] In FIG. **7B**, the main body **8** is further provided with a valve mechanism **89** in the channel **862F**. The valve mechanism **89** has a rubber part **891**, a supporting member **892**, a valve body **893** and a coil spring **894**.

[0085] The rubber part **891** has a bottomless cylindrical shape, and is inserted into the channel **862F** so as to be coaxial with the axis **Ax2**. At the time of insertion, an outer circumferential surface and one end surface of the rubber part **891** make tight contact with the internal surface of the small diameter part **862**. An inner circumferential surface of the rubber part **891** has a diameter which is substantially same as that of the outflow port **862E**, except for the other end of the inner circumferential surface of the rubber part **891**. The other end in the inner circumferential surface of the rubber part **891** projects slightly in both of the inward orientation **r21** and the approaching orientation **z22**, thereby making the diameter of the other end of the inner circumferential surface of the rubber part **891** to be slightly smaller than the outflow port **862E** and/or the needle **44** (see FIGS. **4A** and **4B**). A size in the axial direction **z2** of the rubber part **891** is shorter than that of the neck part **86**.

[0086] The supporting member **892** is, for example, an integrated molded item formed of a resin, and is attached to the channel **862F** so as to bring the rubber part **891** into tight contact with the small diameter part **862**. The supporting member **892** has four side parts **892A** and a bottom part **892B**. For the sake of convenience, only three side parts **892A** are depicted in FIG. **7B**. Each of the side parts **892A** is fixed in the inner circumferential surface of the small diameter part **862**, at a position in the approaching orientation **z22** with respect to the rubber part **891**. A forward end of each of the side parts **892A** makes contact with the other end surface of the rubber part **891**. The respective side parts **892A** are arranged side by side with equal spacing distances in the circumferential direction **62**; each of the side parts **892A** extends from a forward end thereof toward the storing chamber **87**, along the inner circumferential surface of the small diameter part **862**. As appreciated from FIG. **9B**, the bottom part **892B** is cross-shaped in the second plan view, and extends from a position which is apart from the other end surface of the rubber part **891** in the approaching orientation **z22** and which is in the vicinity of the axis **Ax2**, radially toward the ends in

the approaching orientation **z22**, respectively, of the side parts **892A**, and is linked or connected to the ends of the respective side parts **892A**. The respective side parts **892A** and the bottom part **892B** of the supporting member **892** define or demarcates an accommodating space. The accommodating space is substantially cylindrical shaped, and accommodates the valve body **893** and the coil spring **894** therein.

[0087] The valve body **893** and the coil spring **894** are accommodated or stored in the accommodating space of the supporting member **892**. The valve body **893** is accommodated to be movable in the axial direction **z2** in the inside of the accommodating space. The valve body **893** has a circular shape in the second plan view, and has a diameter which is substantially same as the accommodating space having the cylindrical shape. The coil spring **894** is a torsion coil spring, and is positioned between the bottom part of the supporting member **892** and the valve body **893** in the accommodating space. The coil spring **894** makes contact with the valve body **893** in the inside of the accommodating space, and urges the valve body **893** in the separating orientation **z21**. With this, in a case that a contact force in the approaching orientation **z22** is not applied from the needle **44** to the valve body **893**, the valve body **893** makes tight contact with the other end surface of the rubber part **891**, which in turn prevents the ink in the storing chamber **87** from leaking from the outflow port **862E**.

<Liquid Containers **200b** to **200d**>

[0088] The liquid containers **200B** to **200D** are similar to the liquid container **200A**, except for the following points. In each of the liquid containers **200B** to **200D**, the key member is constructed by a combination of at least one kind or two kinds of a first rib, a second rib and a third rib which are similar to the first rib **881**, the second rib **882**, the third rib **883**, respectively. Here, the combinations of the respective ribs including the first, second and third ribs are mutually different among the liquid containers **200A** to **200D**. The three-dimensional shapes of the respective key members are mutually different among the liquid containers **200B** to **200D**, and are also different from the three-dimensional shape of the key member **88**. The three-dimensional shape of the key member is a shape which is defined by the sizes and/or the positions in the axial direction **z2**, the circumferential direction **θ2** and the radial direction **r2** of the respective ribs or by the number (quantity) of the respective ribs. Other than this, the liquid containers **200B**, **200C** and **200D** are different from the liquid container **200A** in view of a point that the liquid containers **200B** to **200C** store, respectively, the inks of the C color, the M color and the Y color. It is also allowable that the liquid containers **200B** to **200D** are different from the liquid container **200A** in view of the ink capacity (an amount of the ink storable therein).

<Container Cap **9**>

[0089] As apparent from FIG. 5B and FIGS. 6A and 6B, the container cap **9** is a single member, and is attachable and detachable with respect to the main body **8**. In the following, unless specifically noted, the term “container cap **9**” means the container cap **9** attached to the main body **8**. In FIG. 8B, the container cap **9** is provided with a ceiling wall **91**, a side wall **92** and a female screw **93**.

<Ceiling Wall **91**>

[0090] As depicted in FIG. 5B, the ceiling wall **91** is a wall having a substantially disc-shape which is coaxial with the axis **Ax1**. In FIG. 8B, the ceiling wall **91** has two main surfaces which are an outer main surface **911** and an inner main surface **912** which are apart from each other in the axial direction **z2**. The inner main surface **912** is positioned in the approaching orientation **z22** with respect to the outer main surface **911**.

<Engaging Part **913** (an Example of a “Sealing Part”), Engaging Part **914** (an Example of a “Circular Contacting Part”>

[0091] An engaging part **913** projects, in the inner main surface **912** of the ceiling wall **91**, in the approaching orientation **z22** at a location close to the axis **Ax1**. The engaging part **913** has a substantially ring shape in the second plan view. The engaging part **913** makes contact liquid-

tightly with the circular flat surface **862C** of the main body **8**, over the entire area thereof in the circumferential direction **62**. With this, in an attached state of the container cap **9**, the engaging part **913** seals, together with the ceiling wall **91**, the outflow port **862E**.

[0092] An engaging part **914** projects, in the inner main surface **912** of the ceiling wall **91**, in the approaching orientation **z22** at a location close to the side wall **92** than the engaging part **913**. The engaging part **914** has a substantially ring shape in the second plan view. The engaging part **914** makes contact liquid-tightly with the end surface **852** of the circular wall **85**, over the entire area thereof in the circumferential direction **62**. With this, in the attached state of the container cap **9**, the engaging part **914** seals, together with the ceiling wall **91**, the opening of the cylindrical space **86A** (see FIGS. **6A** and **6B**).

[0093] It is allowable that the engaging parts **913** and **914** are produced of a same material as that of the ceiling wall **91**, and are integrated with the ceiling wall **91**. The present disclosure is not limited to this, and the engaging parts **913** and **914** may be formed of a material having more flexibility than that of the ceiling wall **91**, for example, of rubber, elastomer, etc., and may be a separate body from the ceiling wall **91**.

<Side Wall **92**, Female Screw **93**>

[0094] As depicted in FIG. **8B**, the side wall **92** is a wall having a substantially cylindrical shape which extends in the approaching orientation **z22** from the outer edge of the inner main surface **912**, and has an inner circumferential surface **921** and an outer circumferential surface **922** which are apart from each other in the radial direction **r1**. The inner circumferential surface **921** has a diameter which is greater to some extent than the outer circumferential surface **853** (see FIG. **6B**) of the circular wall **85**. The female screw **93** is formed in the inner circumferential surface **921** and is capable of being screwed (threadedly engaged) with respect to the male screw **814** of the main body **8**. The container cap **9** is placed to the circular wall **85** from thereabove to cover the circular wall **85** to thereby position the side wall **92** in the outer orientation **r22** with respect to the outer circumferential surface **853**. In a case that the male screw **854** is being screwed with respect to the female screw **93** by rotating the container cap **9** in the circumferential direction **62** in this state, an end, of the side wall **92**, in the approaching orientation **z22** makes contact with the upper surface **841** of the base part **84**, in the entire circumference thereof. A state that the end of the side wall **92** substantially makes contact with the upper surface **841** is an attached state in which the container cap **9** is attached to the main body **8**. By allowing the female screw **93** of the container cap **9** to be screwed with respect to the male screw **854** formed in the main body **8**, the container cap **9** is attached to the main body **8** easily and in an ensured manner. Further, even in a case that the liquid container **200A** is dropped, the container cap **9** is less likely to be detached from the main body **8**.

[0095] After screwing the male screw **854** with respect to the female screw **93**, an end in the approaching orientation **z22** of the side wall **92** (namely, an extending end of the side wall **92**) substantially makes contact with the upper surface **841** over the entire circumference thereof, and is stopped in the axial direction **z2**. The size in the axial direction **z2** of the side wall **92** is determined so that the extending end of the side wall **92** makes contact with the upper surface **841** in the attached state. At least the shape of the engaging part **913** and the size in the axis direction **z2** and the size in the radial direction **r2** of the engaging part **913** are determined so that in the attached state, the engaging part **913** make contact with the circular flat surface **862C**. At least the shape of the engaging part **914** and the size in the axis direction **z2** and the size in the radial direction **r2** of the engaging part **914** are determined so that in the attached state, the engaging part **914** make contact with the end surface **852** of the circular wall **85**.

<Replenishment of Tank **4A** with Ink from Liquid Container **200A**>

[0096] In a case that the tank **4A** is replenished with the ink inside the liquid container **200A**, an operator moves the casing cover **2** in the MFP**100** from the shielding position **P11** to the exposing position **P12** (see FIGS. **1A** and **1i**), moves the tank cover **52A** from the covering position **P31** (see FIG. **1B**) to the exposing position **P32** (see FIG. **3A**), and moves the cap **6A** from the closing

position **P21** (see FIG. 3A) to the opening position **P22** (see FIG. 5A). The operator detaches the container cap **9** in the liquid container **200A** from the main body **8** (see FIGS. 5B, 6A).

[0097] Next, as understood from FIGS. 9A and 9A, the operator turns the outflow port **862E** of the liquid container **200A** to be oriented downward and causes the key member **88** (see FIGS. 6A and 6B) to approach closely to the key hole **48** (see FIGS. 4A and 4B) provided on the MFP **100**. Even in a case that the outflow port **862E** is oriented downward, the valve body **893** makes tight contact with the rubber part **891** by the urging force of the coil spring **894**, and thus the ink inside the storing chamber **87** does not leak out to the outside of the liquid container **200**.

[0098] Next, the operator positions the key member **88** with respect to the key hole **48**. By the positioning, the end surface **852** of the circular wall **85** is positioned immediately above the external space **46E**. Further, the lower end of the first rib **881A** is positioned immediately above the gap defined by the ribs **464** and **465**, the lower end of the first rib **881B** is positioned immediately above the gap defined by the ribs **462** and **463**, and the lower end of the first rib **881C** is positioned immediately above the cutout **468A**. The lower end of the second rib **882A** is positioned immediately above the partially circular space **46D**, and the third rib **883A** is positioned, in the external space **46E**, immediately above the part thereof between the ribs **462** and **4610**. The outflow port **862E** of the liquid container **200A** is positioned immediately above the upper end of the needle **44** of the tank **4A**.

[0099] In a process up to the completion of the positioning, the operator turns the recessed parts **855A** and **855B** of the liquid container **200** leftward and rightward. With this, the operator is capable of positioning the lower ends of the first ribs **881A** and **881B** substantially immediately above the ribs **462** to **465**, even if the operator is unable to visually recognize the key member **88**. With this, the positioning can be performed easily.

[0100] After completing the positioning, the operator moves the key member **88** of the liquid container **200A** downward, in the inside of the key hole **48**. Specifically, the first rib **881A** moves downward between the ribs **464** and **465**, and the first rib **881B** moves downward between the ribs **462** and **463**. Apart in the vicinity of the outer edge of the first rib **881C** is moves downward into the inside of the cutout **468A**. The second rib **882A** moves downward into the inside of the partially circular space **46D**. The third rib **883A** moves downward, in the external space **46E**, to the part thereof between the ribs **462** and **4610**. The outer circumferential surface **862A** moves downward while making contact with the inner ends, respectively, in the ribs **462** to **466** and the inner circumferential surface of the rib **467**.

[0101] There is such a case that the operator is unable to move the key member **88** downward inside the key hole **48**. As one of the factors of such a case, the lower end of the first rib **881A** is positioned erroneously immediately above the space defined by the ribs **462** and **463**, and the lower end of the first rib **881B** is positioned erroneously immediately above the space defined by the ribs **464** and **465**. In this case, since the key member **88** does not move downward inside the key hole **48** due to such a reason, for example, that the lower end of the first rib **881C** interferes with the upper end of the rib **469**, etc., the operator rotates the liquid container **200A** by 180 degrees in the circumferential direction $\theta 2$, and positions the lower ends, respectively, of the first ribs **881A** and **881B** to the correct positions thereof, respectively. Afterwards, the operator moves the key member **88** downward in the inside of the key hole **48**.

[0102] In the process in which the key member **88** is (being) moved downward inside the key hole **48**, the needle **44** is inserted from the outflow port **862E** of the liquid container **200A** to the channel **862F**, and approaches toward the valve body **893**. After the upper end of the needle **44** makes contact with the lower end of the valve body **893**, the valve body **893** starts moving upward, by the contact force received from the upper end of the partition wall **443** of the needle **44**, against the urging force of the coil spring **894**. In response to a situation that the circular flat surface **862C** enters into the circular space **48B**, and the circular flat surface **862B** makes contact with the contact surface **461A**, and the end surface **852** of the circular wall **85** reaches the upper surface **43** in the

external space **46E**, the key member **88** of the tank **4** is engaged to the key hole **48** of the main body **8**, thereby completing the connection of the main body **8** to the tank **4A**.

[0103] At the time of completion of the connection, the contact surface **461A** makes contact with the circular flat surface **862B** of the small diameter part **862** which is positioned in the cylindrical space **46A**. The inner ends, respectively, of the ribs **462** to **466** and the inner circumferential surface of the rib **467** make contact, over the entire area thereof in the up-down direction **z1**, with the outer circumferential surface **862A** of the small diameter part **862** which is in the inside of the cylindrical space **46A**. The rib **467** further makes contact with the inner circumferential surface of the third rib **883A** from the side of the inward orientation **r21**. At the time of completion of the connection (an example of a “connected state”), the end surface **852** of the circular wall **85** makes contact with the upper surface **43** at a position which is in the inner orientation **r21** with respect to the projecting wall **45**. Owing to this, even in a case that the operator removes his or her hand from the main body **8**, the main body **8** is supported by the upper surface **43**, the projecting wall **45** and the receiver key member **46** of the key hole **48** as depicted in FIGS. **9A** and **9B**, and stands in an inverted manner, without hardly being inclined with respect to the up-down direction **z1**.

[0104] At the time of completion of the connection, the valve body **893** releases the channel **862F** by the contact force from the forward end of the partition wall **443**. Since the forward end of the partition wall **443** projects to a location above the upper ends, respectively, of the channels **441** and **442**, gaps are defined each between the valve body **893** and the channel **441** and between the valve body **893** and the channel **442**. With these gaps, the storing chamber **87** of the main body **8** and the storing chamber **47** of the tank **4A** communicate with each other via the channels **441**, **442** and **862F**. Namely, the main body **8** and the tank **4A** are connected so as to allow the ink to flow out from the storing chamber **87** to the storing chamber **47**.

[0105] There is such a case that the ink adheres to the surface of the neck part **86**, etc. In a process that the neck part **86** approaches toward the contact surface **461A** in the inside of the cylindrical space **46A**, the ink leaking out from a location between the neck part **86** and the contact surface **461A** flows into the partially circular spaces **46C** and **46D** and/or flows into the external space **46E** via the gap between the ribs **462** and **463** and the gap between the ribs **464** and **465**. Further, the ink overflowed from the partially circular spaces **46C** and **46D** flows out to the external space **46E** via the cutouts **486A** and **469A**.

[0106] Since immediately after the completion of connection, gas-liquid replacement starts between the liquid container **200A** and the tank **4A**. In the gas-liquid replacement, the ink inside the storing chamber **87** flows into the storing chamber **47** via the channels **862F** and **441**. In the gas-liquid replacement, air flows from the atmosphere communicating hole of the tank **4A** into the storing chamber **47**, and this air flows into the storing chamber **87** via the channels **442** and **862F**. An outflow amount of the ink from the storing chamber **87** to the storing chamber **47**, and an inflow amount of the air from the storing chamber **47** to the storing chamber **87** are substantially same. In a case that the liquid surface of the ink in the storing chamber **47** reaches the lower end of the flow channel **442** or that the ink inside the storing chamber **87** becomes empty, the gas-liquid replacement is ended. In such a manner, the tank **4A** is replenished with the ink inside the liquid container **200A**.

[0107] After the ending of the ink replenishment, the operator pulls the key member **88** and the neck part **86** of the liquid container **200A** upward from the key hole **48** and the needle **44** of the tank **4A**. In a process in which the neck part **86** is (being) moved upward with respect to the needle **44**, the valve body **893** first maintains a state that the valve body **893** makes contact with the upper end of the partition wall **443** of the needle **44** by the urging force of the coil spring **894**. After the valve body **893** makes contact with the small diameter part **862** of the neck part **86**, the valve body **893** is apart from the upper end of the circular wall **443**.

[0108] Afterward, the operator attaches the container cap **9** to the main body **8** (see FIGS. **5B** and **6A**). With this, the ink remaining in the inside of the storing chamber **87** of the liquid container

200A is preserved. The operator moves the casing cover **2** in the MFP **100** from the exposing position **P12** to the shielding position **P11** (see FIGS. **1A**, **1**), moves the tank cover **52A** from the exposing position **P32** (see FIG. **3A**) to the covering position **P31** (see FIG. **1**), and moves the cap **6A** from the opening position **P22** (see FIG. **5A**) to the closing position **P21** (see FIG. **3A**). In a case that the tank **4A** is replenished with the ink inside of the liquid container **200A** in another occasion, the operator replenishes the tank **4A** with the ink in a procedure which is similar to that described above.

<Prevention of Erroneous Connection of Liquid containers **200B** to **200D** to Tank **4A**>

[0109] In a case that the operator attempts to erroneously connect any one of the liquid containers **200B** to **200D** to the key hole **48** of the tank **4A**, the shape and/or the position in the left-right direction **x1** and/or the front-rear direction of at least one of the ribs **462** to **4610** constructing the key hole **48** does or do not match the key member of each of the liquid containers **200B** to **200D** which is wrong (not matching) with respect to the tank **4A**, in some cases, which in turn results in such a case that the key member is not in conformity to the key hole **48**. Owing to this, the operator can quickly recognize that the operator is attempting to fit any one of the liquid containers **200B** to **200D** erroneously to the tank **4A**. Further, it is also possible to quickly prevent the tank **4A** from being replenished with an ink which is wrong with respect to the tank **4A**.

[0110] Furthermore, in such a case that the operator attempts to erroneously fit any one of the liquid containers **200B** to **200D** to the key hole **48** of the tank **4A**, the shape and/or the position in the up-down direction **z1** of at least one of the ribs **462** to **4610** constructing the key hole **48** does or do not match the key member of each of the liquid containers **200B** to **200D** which is wrong (not matching) with respect to the tank **4A**, in some cases. In this case, since any one of the liquid containers **200B** to **200D** does not move downward, the operator can quickly recognize that the operator is attempting to fit any one of the liquid containers **200B** to **200D** erroneously to the tank **4A**. Moreover, it is also possible to prevent the tank **4A** from being replenished with an ink which is wrong with respect to the tank **4A**.

Technical Effects of Embodiment

[0111] In the ink replenishment, the outflow port **862E** of the main body **8** is oriented downward and thus the ink easily adheres to the surrounding of the outflow port **862E**; thus, in a process of making the liquid container **200A** to be in the placement position after the ink replenishment, this ink remains, along an outer surface of the neck part **86**, between the circular wall **85** and the neck part **86**. With this, during a process of replenishing the tank **4A** with the ink again, the ink is less likely to adhere to a hand and/or a finger of the operator and/or to a place in which the liquid container **200A** is placed (for example, the desktop, etc.). Further, in the attached state wherein the container cap **9** is attached to the main body **8**, the engaging part **913** seals the outflow port **862E** and thus the ink remaining between the circular wall **85** and the neck part **86** does not flow back to the outflow port **862E**. Furthermore, in the attached state, the opening of the cylindrical space **86A** (see FIG. **6B**) is sealed by the engaging part **914**, and thus the ink inside the cylindrical space **86A** is prevented from flowing down or streaming along the outer circumferential surface **853** of the circular wall **85** and from adhering to the outer surfaces, respectively, of the base part **84**, the shoulder part **83** and the trunk part **82**. Further, the ink is prevented from adhering to the side wall **92** of the bottle cap. With this, the ink is less likely to adhere to the hand and/or the finger of the operator and/or to the place wherein the liquid container **200A** is placed.

[0112] By performing the cutting out in the male screw **854**, the recessed parts **855A** and **855B** are formed at two locations, respectively, of the outer circumferential surface **853**. The recessed parts **855A** and **855B** are used, by the operator in the positioning in the ink replenishment, as the indicators indicating the positions of the first ribs **881A** and **881B**, respectively, each of which is a part of the key member **88**. Owing to such recessed parts **855A** and **855B**, the operator is capable of easily position the key member **88** with respect to the key hole **48**.

[0113] The key member **88** is formed in the cylindrical space **86A**, and is not formed on the outer

circumferential surface **853** of the circular wall **85**. Such an outer circumferential surface **853** is formed with the male screw **854**. Accordingly, the key member **88** does not affect the screwing (threadedly engagement) of the male screw **854** and the female screw **93**. With this, the container cap **9** is attached to the main body **8** easily and in an ensured manner.

[0114] The key member **88** is formed in the cylindrical space **86A**, and is loosely in conformity to the key hole **48** formed in the tank **4A**. The number (quantity), the three-dimensional shape and/or the position of the ribs constructing the key member **88** and the key hole **48** are changed per each of the liquid containers **200A** and **200D**. Accordingly, the operator is capable of grasping the kind of the liquid containers **200A** to **200D** by the shape of the key member **88**.

[0115] In the ink replenishment, the ink adhered to the neck part **86** easily drips down from the neck part **86**. The forward end of the circular wall **85**, however, projects in the separating orientation **z21** with respect to the forward end of the neck part **86**. Accordingly, even in a case that the ink drips down when the main body **8** is inclined so that the orientation of the outflow port **862E** is reversed, the ink is easily made to drip down to the inside of the circular wall **85**. Further, in a case, for example, that the main body **8** is dropped down from the desk, etc., the circular wall **85** collides against the floor, etc., faster than the neck part **86**, and thus the neck part **86** can be protected from the impact.

[0116] The operator is capable of replenishing the tank **4A** with the ink by using the liquid container **200A** a plurality of times. Each time the ink replenishment is performed, as appreciated from FIG. **9B**, since the outflow port **862E** of the main body **8** is oriented downward, the ink easily adheres to the surrounding of the outflow port **862E**. In the process of making the liquid container **200A** to be in the placement position after the ink replenishment, the ink flows down the outer circumferential surface of the neck part **86** and remains in the cylindrical space **86A** between the circular wall **85** and the neck part **86**. In the next ink replenishment, in a case that the outflow port **862E** is oriented downward, the ink inside the cylindrical space **86A** moves downward toward the upper surface **43** of the tank **4A**. However, at the time of completion of the connection, the end surface **852** of the circular wall **85** makes contact with a position, in the upper surface **43**, in the inner orientation **r21** with respect to the projecting wall **45**. Accordingly, the ink remains in the external space **46E**, does not leak from the projecting wall **45** in the outward orientation **r22** in the upper surface **43**, and does not flow down the outer surface of the tank **4A**, including the upper surface **43**.

[0117] As depicted in FIG. **4B**, the distance **Dz1** of the projecting wall **45** and the distances **Dz2** and **Dz3** in the circular wall **85** are in a relationship of $Dz2 > Dz3 > Dz1$. Under this relationship, by making a distance in the left-right direction **x1** between the connecting plates **73** and **74** to be longer to some extent than the diameter of the outer circumferential surface **853** (see FIG. **6B**) of the circular wall **85**, it is possible to make the three-dimensional shape of the projecting wall **45** to be small, without any interference between the projecting wall **45** and the male screw **854**.

[0118] While the present disclosure has been described in conjunction with various example structures outlined above and illustrated in the figures, various alternatives, modifications, variations, improvements, and/or substantial equivalents, whether known or that may be presently unforeseen, may become apparent to those having at least ordinary skill in the art. Accordingly, the example embodiments of the disclosure, as set forth above, are intended to be illustrative of the invention, and not limiting the invention. Various changes may be made without departing from the spirit and scope of the disclosure. Therefore, the disclosure is intended to embrace all known or later developed alternatives, modifications, variations, improvements, and/or substantial equivalents. Some specific examples of potential alternatives, modifications, or variations in the described invention are provided below:

Modification

[0119] In the following, a body **41**, a main body **8** and a container cap **9** according to a modification will be explained, with reference to FIGS. **10A**, **10B**, **11A**, **11B** and **12**. In the

following, the difference from the embodiment will be mainly explained, a configuration which corresponds to that explained in the embodiment is denoted by a same reference numeral, and any explanation therefor will be omitted or simplified.

<Body **41**, Curved Plates **71** and **72**, Connecting Plates **73** and **74**>

[0120] In FIG. **10A**, the upper ends of the curved plates **71** and **72** and the connecting plates **73** and **74** are different from those in the embodiment in view of that the upper ends of the curved plates **71** and **72** and the connecting plates **73** and **74** are positioned above the upper end of the needle **44**.

[0121] The connecting plate **74** is different from that of the embodiment in view of that a rib **741**, a groove **742** and a slit **743** are arranged side by side from the front to the rear in this order.

[0122] The rib **741** perpendicularly projects to the left side, in the inner surface of the connecting plate **74** at a location in front of the needle **44**. The rib **741** is continuous between both ends in the up-down direction **z1** of the connecting plate **74**, and extends linearly in the up-down direction **z1**. The rib **741** has a rectangular plate-like shape which is thin in the front-rear direction **y1** and elongated in the up-down direction **z1** in a first front view (which is in a plan view from the front).

[0123] The groove **742** is positioned on the right side as seen from the needle **44**, is continuous between the both ends in the up-down direction **z1** of the connecting plate **74**, and extends linearly in the up-down direction **z1**. The groove **742** is recessed rightward from the inner surface of the connecting plate **71**. The depth and/or the width of the groove **742** is/are substantially constant in an entire area, of the groove **742**, between the both ends in the up-down direction **z1** of the groove **742**.

[0124] The slit **743** is continuous from a position, in the connecting plate **74**, which is above to some extent from a lower end of the connecting plate **74** and the upper end of the connecting plate **74**, and extends linearly in the up-down direction **z1**. A distance between the both ends in the up-down direction **z1** of the slit **743** is substantially same as a distance between both ends in the axial direction **z2** of a rib **813** (see FIGS. **11A** and **11B**). In the following, the distance between both ends in the up-down direction **z1**, and the distance between both ends in the axial direction **z2** are each referred also simply as a “height”. The width of the slit **743** is substantially constant over an entire area between the both ends in the up-down direction **z1** of the slit **743**.

[0125] In FIGS. **10A** and **10B**, the connecting plate **73** is different from that in the embodiment in view of that the connecting plate **73** is constructed of an inner plate **75** and an outer plate **76** which expand in the up-down direction **z1** and the front-rear direction **y1**. The inner plate **75** faces the outer plate **76** in the left-right direction **x1**, at a position which is closer to the needle **44** to some extent than the outer plate **76**. A slit **731** and a rib **732** are formed in the inner plate **75**.

[0126] The slit **731** is positioned on the left side as seen from the needle **44**, is continuous between a position which is to some extent above a lower end of the inner plate **75** and an upper end of the inner plate **75**, and linearly extends in the up-down direction **z1**. The slit **731** has a substantially same height as a rib **815** (see FIGS. **11A** and **11B**). The width of the slit **731** is substantially constant in an entire area between both ends in the up-down direction **z1** of the slit **731**.

[0127] The rib **732** perpendicularly projects to the right side, in the inner surface of the inner plate **75** at a location behind the needle **44**. The rib **732** is continuous between the upper end and the lower end of the inner plate **75**, and extends linearly in the up-down direction **z1**. The rib **732** has a plate-like shape which is similar to that of the rib **741** of the connecting plate **74**. A projecting end surface (namely, a right side surface) of the rib **732** expands in the up-down direction **z1** and the front-rear direction **y1**, and is inclined to some extent with respect to the left-right direction **x1**.

[0128] In FIG. **10A**, the curved plate **71** is different from that of the embodiment in view of that a projecting part **711** is formed in the curved plate **71**. The projecting part **711** is positioned apart from the needle **44** at a location which is substantially in front of the needle **44**, and is positioned in the vicinity of the center in the circumferential direction $\theta 1$ of the curved plate **71**. The projecting part **711** has a substantially rectangular shape in the first plan view, and projects rearward from the inner surface of the curved plate **71**. A rear end surface of the projecting part **711** is a circular arc

shape in the first plan view (see FIG. 10B). A right side surface and a left side surface of the projecting part **711** are each a flat surface which is substantially orthogonal to the left-right direction $x1$. The projecting part **711** is continuous between the both ends in the up-down direction $z1$ of the curved plate **71**, except for a part **712** which is on a rear right corner in the first plan view of the projecting part **711**, and extends linearly in the up-down direction $z1$. The part **712** is a substantially parallelogram in the first plan view, is continuous between a lower end of the curved plate **71** and a location which is below the upper end of the curved plate **71**, and extends in the up-down direction $z1$.

[0129] The projecting wall **45** partitions or defines a key hole **48** which is released upward, together with the ribs **741** and **732**, the groove **742**, the slits **743** and **731**, the projecting part **711** and the part **712**. The key hole **48** is an example of a “receiver key member” and/or an example of a “tank recessed part”. A liquid container **200A**, which is to be described later on, is connected to the key hole **48** in a case of ink replenishment. Although the key hole **48** is in conformity to a key member **88** formed on the side of the liquid container **200A**, the key part **48** is not in conformity to a key member of each of other liquid containers **200B** to **200D**.

<Circular Wall **85**>

[0130] In FIGS. **11A** and **11B**, the circular wall **85** is different from that of the embodiment in view of having connecting plates **856** and **857** and curved plates **858** and **859**. The circular wall **85** is different from that of the embodiment also in view of that an end surface in the separating orientation $z21$ of the circular wall **85** does not have a ring shape (annular shape).

[0131] Each of the connecting plates **856** and **857** is a flat plate having a substantially linear shape in the second plan view, and the connecting plates **856** and **857** face each other in the radial direction $r2$ with the neck part **86** being interposed therebetween. At the time of ink replenishment, the connecting plate **856** is positioned on the left side with respect to the connecting plate **74** of the projecting wall **45** (see FIG. **10A**), and the connecting plate **857** is positioned on the right side with respect to the connecting plate **73** of the projecting wall **45** (see FIGS. **10A** and **10B**).

[0132] Each of the curved plates **858** and **859** has a circular arc shape located on a virtual circle $c2$ (see FIG. **11B**) in the second plan view, and the curved plates **858** and **859** face each other in the radial direction $r2$ with the neck part **86** being interposed therebetween. The virtual circle $c2$ is a circle having the axis $Ax2$ as the center thereof in the second plan view, and has a diameter which is smaller to some extent than that of the upper surface **841** of the base part **84**. Specifically, the planar shape of the upper surface **841** of the base part **84** is substantially circular. The curved plate **858** extends, in the upper surface **841**, in the separating orientation $z21$ from a position along the virtual circle $c2$. The curved plate **859** extends, on the upper surface **841**, in the axial direction $z2$ from a position which is rotationally moved approximately by 180 degrees, from the position occupied by the curved plate **858** on the upper surface **841**, in a circumferential direction $\theta2$ of the axis $Ax2$. The curved plate **858** is connected or linked to one ends, respectively, of the connecting plates **856** and **857**, which are located to approach toward each other in the circumferential direction $\theta2$. The other ends, respectively, of the connecting plates **856** and **857** are connected to each other by the curved plate **859**. The curved plate **858** has such a shape that, at the time of ink replenishment, the curved plate **858** is overlapped with the curved plate **71** of the projecting wall **45** at a position immediately behind the curved plate **71** (see FIGS. **10A** and **10B**) of the projecting wall **45**; the curved plate **859** has such a shape that, at the time of ink replenishment, the curved plate **859** is overlapped with the curved plate **72** of the projecting wall **45** (see FIGS. **10A** and **10B**) of the projecting wall at a position immediately in front of curved plate **72**.

[0133] In FIGS. **11A** and **11B**, ribs **811** to **813** project perpendicularly from the outer surface of the connecting plate **856** toward the outside of the circular wall **85**. Among the ribs **811** to **813**, the rib **811** is positioned closest to the curved plate **858** and the rib **813** is positioned closest to the curved plate **859**. The ribs **811** to **813** are continuous at both ends in the axial direction $z2$ of the connecting plate **856**, and extend linearly in the axial direction $z2$. Each of the ribs **811** to **813** has a

substantially same height as the slit **743** (see FIG. **10A**) in the tank **4A**. The ribs **811** to **813** do not protrude, in the second plan view, from the upper surface **841** of the base part **84**. Specifically, the maximum distance between the both ends in the projecting direction of the rib **811** is shorter than the distance between the both ends in the projecting direction of the rib **741** (see FIG. **10A**) on the side of the tank **4A**. In a case of comparing a distance from the axis **Ax2** to the projecting end of the rib **811** in the radial direction **r2** with a distance from the axis **Ax2** to the outer circumferential surface of the curved plate **858** in the radial direction **r2**, the distance to the projecting end (an example of a “first distance”) is shorter than the distance to the outer circumferential surface (an example of a “second distance”) in order to avoid any interference with respect to the container cap **9**. At a time of completing the connection, the rib **811** makes contact with the rib **741** on the side of the tank **4A** (see FIG. **10A**) from the rear side. In this situation, the projecting end (namely, the left end) of the rib **741** makes contact with the connecting plate **856** from the right side. The rib **812** is engaged to the groove **742** at a time of connecting the bottom **200A** with the tank **4A**; in this situation, the projecting end of the rib **812** makes contact with the bottom of the groove **742** from the left side. The rib **813** is inserted into the slit **743** at the time of completion of connection of the bottom **200A** with the tank **4A**. A part of the projecting end surface of the rib **813** is notched and forms a part of a male screw **814** (see FIG. **11A**) which will be described later on.

[0134] In FIGS. **11A** and **11B**, a rib **815** and a groove **816** are formed on the outer surface of the connecting plate **857**. The rib **815** projects perpendicularly and toward outside of the circular wall **85** in the outer surface of the connecting plate **857**, at a location closer to the curved plate **858** than the groove **816**. A projecting end of the rib **815** does not protrude, in the second plan view, from the upper surface **841** in the outward orientation **r22**. The rib **815** has a height which is substantially same as the height of the slit **731** (see FIGS. **10A** and **10B**). Specifically, ends in the approaching orientation **z22**, respectively, of the rib **815** and the connecting plate **857** are located at mutually same positions. On the other hand, an end in the separating orientation **z21** of the rib **815** is at a location which is closer, to some extent, to the approaching orientation **z22** than an end in the separating orientation **z21** of the connecting plate **857**, as depicted in FIG. **11A**. Namely, a distance in the axial direction **z2** from the upper surface **841** to the end in the separating orientation **z21** of the rib **815** is shorter than a distance in the axial direction **z2** from the upper surface **841** to the end in the axial direction **z2** of the connecting plate **857**. Note that the separating orientation **z21** is the up-down direction **z1** in the placement posture, and the end in the separating orientation **z21** is the upper end in the placement posture. At the time of completing the connection, the rib **815** is inserted into the slit **731**, and the end in the separating orientation **z21** of the rib **815** makes contact with a lower end of the slit **731**. The groove **816** is continuous between the both ends in the axial direction **z2** of the connecting plate **857**, and extends linearly in the axial direction **z2**. The groove **816** is recessed from the outer surface to the inner surface of the connecting plate **857**. The bottom surface of the groove **816** is parallel to the circumferential direction **θ2** in the first plan view. The width of the groove **816** is substantially constant between both ends in the axial direction of the connecting plate **857**. At the time of the ink replenishment, the rib **732** (see FIGS. **10A** and **10B**) is engaged to the groove **816**; in this situation, the projecting end of the rib **732** makes contact with the bottom of the groove **816** from the left side.

[0135] In FIGS. **11A** and **11B**, another part of the male screw **814** is formed in the outer circumferential surface of each of the curved plates **858** and **859**, at a location in the vicinity of the center in the axial direction **z2**. As described above, the part of the male screw **814** is formed also in the rib **813**. Namely, the male screw **814** is formed in a divided manner in the projecting end surface of the rib **813** and the curved plates **858** and **859**. Such a male screw **814** is screwed (threadedly engaged) with respect to a female screw **93** formed in the container cap **9**.

[0136] A groove **817** is formed in the outer surface of the curved plate **858**. The groove **817** is continuous between both ends in the axial direction **z2** of the curved plate **858**, at a location in a center part in the circumferential direction **θ1** of the curved plate **858**, and extends linearly in the

axial direction **z2**. The groove **817** is recessed from the outer circumferential surface toward the inner circumferential surface of the curved plate **858**. A bottom surface of the groove **817** is parallel to the circumferential direction $\theta 2$ in the first plan view. The width and the depth of the groove **817** are substantially constant between the both ends in the axial direction **z2** of the curved plate **858**, except for a part corresponding to a rib **818** which will be described later on. Specifically, the depth of the groove **817** is substantially same as a size in the front-rear direction **y1** of the projecting part **711**, and the width of the groove **817** is same as the maximum value of the size in the left-right direction **x1** in the projecting part **711**. A rib **818** is formed in the groove **817**. The rib **818** extends, in the groove **817**, from one side surface in the circumferential direction $\theta 2$ (a side surface in a clockwise orientation in FIGS. **11A** and **11B**) of the groove **817**. The rib **818** extends from a location, in the groove **817**, between both ends in the radial direction **r2** of the groove **817**, and expands in the radiation direction **r2** and the circumferential direction $\theta 2$. The rib **818** has a shape which is substantially same as that of the part **712** (see FIG. **10A**) on the side of the tank **4A**, in the second plan view, and the rib **818** has a plate-like shape which is thin in the axial direction **z2**. An end surface in the separating orientation **z21** of the rib **818** is apart (separated), in the approaching direction **z22**, from an end in the separating orientation **z21** of the groove **817**, by an amount corresponding to the height of the part **712** (see FIG. **10A**).

[0137] The groove **817** is engaged to the projecting part **711** (see FIGS. **10A** and **10B**) at the time of completing the connection; in this situation, the end surface in the separating orientation **z21** of the rib **818** makes contact with the part **712** from thereabove.

[0138] The circular wall **85** forms a key member **88** which is in conformity to the key hole **48** (see FIGS. **10A** and **10B**), together with the ribs **811** to **813**, **815** and **818** and the grooves **816** and **817**. In the embodiment, the key member **88** is positioned between the circular wall **85** and the neck part **86**. In the present modification, however, the key member **88** is positioned at the outer surface of the circular wall **85**, rather than being positioned between the circular wall **85** and the neck part **86**.
<Seat Part **865**, Seat Surface **865A**>

[0139] In FIGS. **11A** and **11B**, the neck part **86** is different from that of the embodiment in view that the neck part **86** is provided with a seat part **865** having a seat surface **865A**. The seat surface **865A** has a substantially ring shape surrounding the entire circumference of the neck part **86** in the second plan view, at the outside of the neck part **86**. The seat surface **865A** is a surface parallel to the radial direction **r2**. Specifically, the seat surface **865A** extends in the outward orientation **r22** toward the circular wall **85** from the entire circumference of the outer circumferential surface, in the neck part **86**, which is apart from the forward end surface in the approaching orientation **z22**. Namely, the seat surface **865A** is at a position which is closer to the circular wall **85** in the radial direction **r2** than the forward end surface of the neck part **86** and which is closer to the upper surface **841** and/or the storing chamber **87** in the axial direction **z2** than the forward end surface of the neck part **86**. Further, in the placement posture of the main body **8**, the seat surface **865A** is oriented upward at a location below the forward end surface of the neck part **86**. Although the seat surface **865A** is continuous with the inner surfaces, respectively, of the connecting plates **856** and **857** and the bottom wall of the groove **817**, the seat surface **865A** is not continuous with the inner surfaces, respectively, of the curved plates **858** and **859**. The width in the radial direction **r2** of the seat surface **865A** becomes to a minimum width **W11** between the neck part **86** and the bottom wall of the groove **817**.

<Container Cap **9**>

[0140] In FIG. **12**, the container cap **9** is different from that of the embodiment in view of further having a circular projecting piece **94**.

[0141] The circular projecting piece **94** is a wall having a substantially cylindrical shape and extending in the approaching orientation **z22** from a position which is in the outward orientation **r22** with respect to the engaging part **913** and which is in the inward orientation **r21** with respect to the side wall **92**. The inner circumferential surface of the circular projecting piece **94** is

substantially coaxial with the outer circumferential surface of the neck part **86**. The thickness of the circular projecting piece **94** is a size in the radial direction $r1$ between the inner circumferential surface and the outer circumferential surface of the circular projecting piece **94**. This thickness is substantially constant over the entire circumference in the circumferential direction $\theta1$, and is smaller slightly than the above-described minimum width $W11$ between the neck part **86** and the groove **817**. In the attached state of the container cap **9**, the circular projecting piece **94** makes contact with the outer circumferential surface of the neck part **86** in the main body **8**, and is fitted between the neck part **86** and the outer circumferential surface of the circular wall **85**.

[0142] In a process of screwing the male screw **814** with the female screw **93** (hereinafter also referred to as a “screwing process”), the inner circumferential surface of the circular projecting piece **94** slidably moves on the outer circumferential surface of the neck part **86**, while rotating about the axis $Ax2$. After the screwing process, an end **941** in the approaching orientation $z22$ (namely, an extending end) of the circular projecting piece **94** makes contact with the seat surface **865A** over the entire circumference thereof. By the contact between the end **941** and the seat surface **865A**, an end position at which the container cap **9** is screwed with respect to the main body **8** is determined, and the screwing of the container cap **9** is stopped in the axial direction $z2$. The size in the axial direction $z2$ of the circular projecting piece **94** is previously determined so that the end **941** makes contact with the seat surface **865A** in the attached state. Further, in a case that the container cap **9** is at the end position, the inner main surface **912** makes contact with an end surface in the separating orientation $z21$ of the circular wall **85**. With this, the container cap **9** seals a space which is on the inner side with respect to the circular wall **85**, in a liquid tight manner.

Other Modifications

[0143] In the embodiment, the four color inks are stored in the tanks **4A** to **4D**, respectively, as described above. It is allowable, however, that a pre-processing liquid (another example of the “liquid”) which is discharged or ejected by the recording head **322** onto a sheet **S**, etc., prior to the discharge of the ink in the image recording is stored in the tanks **4A** to **4D**. Other than this, it is allowable that the tanks **4A** to **4D** store water (yet another example of the “liquid”) which is used for cleaning or washing the recording head **322**.

[0144] In the embodiment, the printer part **3** is capable of recording a full color image on the sheet **S**. The present disclosure, however, is not limited to or restricted by this; the printer part **3** may be capable of recording only a monochrome image on the sheet **S**. In such a case, the tank set **31** is provided with the tank **4A**, the holding member **51A**, the cap **6A** and the tank cover **52A**.

[0145] In the embodiment, the key hole **48** is provided on the tank **4A**. The present disclosure, however, is not limited to this; the key hole **48** may be formed in the inner circumferential surface of the through hole **511A** of the holding member **51A**.

[0146] In the embodiment, the three-dimensional shape of each of the key member **88** and the key hole **48** is made to be mutually different per each of the colors of the inks. The present disclosure, however, is not limited to this; the three-dimensional shape of each of the key member **88** and the key hole **48** is made to be mutually different per each kind (namely, each model) of the MFP **100**.

[0147] In the embodiment, each of the key member **88** and the receiver key member **46** is constructed of the cutout and the rib projecting in the separation orientation $z21$ with respect to the upper surface **841**. The present disclosure, however, is not limited to this; other than this, each of the key member **88** and the receiver key member **46** may be constructed of a slit which is long in the approaching orientation $z22$ with respect to the upper surface **841** or a recessed part which is recessed in the circumferential direction $\theta2$ or the radial direction $r2$.

[0148] In the embodiment, the main body **8** is provided with the recessed parts **855A** and **855B**, and the recessed parts **855A** and **855B** are formed by performing cutting out the parts, in the male screw **854**, which are rotationally moved with each other only by 180 degrees. The present disclosure, however, is not limited to this; the main body **8** may be provided with two projected parts, rather than the recessed parts **855A** and **855B**. The two projected parts may project from

parts, respectively, in the outer circumferential surface **853**, which are rotationally moved with each other only by 180 degrees.

Claims

1. A liquid supplying apparatus comprising: a liquid container having an internal space for storing liquid, the liquid container including: a nozzle; and a first wall located around the nozzle; and a tank having a body defining a chamber for storing the liquid, the body including: a tube connecting the chamber to an outside of the body; and a second wall located around the tube and extending in a direction extending the tube, wherein the tube is configured to be inserted into the liquid container so as to allow the liquid to flow out from the liquid container to the chamber of the tank, and in a case where the liquid container is connected to the tank, the first wall is located between the tube and the second wall.
 2. The liquid supplying apparatus according to claim 1, wherein a first length of the first wall in a direction extending the nozzle is greater than a second length of the second wall in the direction extending the tube.
 3. The liquid supplying apparatus according to claim 2, wherein the first wall has a screw part, the liquid container further includes a cap configured to be screwed with the screw part of the first wall, and a third length of the screw part in the direction extending the nozzle is greater than the second length.
 4. The liquid supplying apparatus according to claim 1, wherein the liquid container further includes a key member, and the liquid supplying apparatus further includes a receiver key member corresponding to the key member.
 5. The liquid supplying apparatus according to claim 4, wherein the tank includes the receiver key member.
 6. The liquid supplying apparatus according to claim 4, further comprising a plurality of tanks including the tank, each of the plurality of tanks corresponding to a color of the liquid to be stored, wherein a position of the receiver key member is different with respect to each of the plurality of tanks, and the key member is positioned so that the key member is engaged to the receiver key member of one of the plurality of tanks corresponding to the key member.
 7. The liquid supplying apparatus according to claim 4, wherein the key member is located in a space between the nozzle and the first wall.
 8. The liquid supplying apparatus according to claim 7, wherein the key member is connected to the nozzle and the first wall.
 9. The liquid supplying apparatus according to claim 7, wherein the key member is connected to the nozzle and is not connected to the first wall.
 10. The liquid supplying apparatus according to claim 7, wherein the key member is not connected to the nozzle and is connected to the first wall.
 11. The liquid supplying apparatus according to claim 7, wherein the key member extends outward from the first wall.
 12. The liquid supplying apparatus according to claim 1, wherein an upper end of the first wall is located above a forward end of the nozzle.
 13. The liquid supplying apparatus according to claim 4, further comprising: a plurality of tanks including the tank, each of the plurality of tanks corresponding to a color of the liquid to be stored; and a plurality of receiver key members including the receiver key member, wherein the plurality of receiver key members have different shapes corresponding to the plurality of tanks, respectively, and the key member has a shape so that the key member is in conformity with the receiver key member of one of the plurality of tanks corresponding to the key member.
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